

Haoling(Brian) Pu

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EDUCATION

Carnegie Mellon University

Master of Science in Artificial Intelligence and Innovation, School of Computer Science (GPA: 4.00/4.00)

Pittsburgh, PA

May 2027

Related Courseworks: AI Engineering, Deep Learning, Multimodal ML, Advanced Natural Language Processing, Generative AI

University of Michigan - Ann Arbor

Bachelor of Science in Computer Science & Data Science (GPA: 3.95/4.00)

Ann Arbor, MI

May 2025

Related Courseworks: Database system, Data Structure and Algorithm, Machine Learning, Web Systems, Computer Organization

SKILLS

Programming Languages: Python, C++, C#, SQL, JavaScript, Golang, Java

LLM & Agent Engineering: Claude, Codex, Agentic Workflows, Tool Calling, RAG, Vector Databases (Chroma), LangChain

ML, Systems & Cloud: PyTorch, CUDA, MLflow, vLLM, Scikit-learn, Slurm, Kubernetes, Docker, AWS

WORK EXPERIENCE

Software Engineering Intern | Google, Los Angeles, CA

05/2026-08/2026

- Engineered a **NotebookLM-style agentic knowledge system** that automatically generates, serves, and continuously self-maintains a team engineering knowledge base for a **20+ engineer team**
- Deployed** the end-to-end agentic pipeline as scheduled distributed jobs, automating recursive source discovery, delta ingestion, **Gemini** synthesis, semantic auditing, and automated change delivery across **1,000+** engineering docs
- Developed the team-wiki system using a **multi-agent engineering workflow(EGM)**, orchestrating AI agents for architecture design, TDD implementation, code review, adversarial design critique, and automated debugging
- Productionized the system for **cross-team adoption at Google**, exposing it through an reusable **AI agent skill** that delivers source-cited answers, accelerating onboarding and knowledge discovery

Research Assistant @ Li Lab | CMU Language Technology Institute, Pittsburgh, PA

10/2025 - Present

- Developed a reference-free, future-consistent **decoding framework** for simultaneous speech translation, using **Gemma-4** and **Qwen3-30B** to model plausible source continuations and gate target emissions through cross-future agreement; reduced premature commitments and reached COMET parity with a reference-based system under low-latency settings
- Engineered a self-hosted training and inference pipeline on **NVIDIA L40S** GPUs, serving Qwen3-30B-A3B-Instruct-FP8 with **vLLM**, generating streaming trajectories from 40K GigaSpeech utterances, **LoRA**-fine-tuning Qwen3-Omni with Megatron-SWIFT, and achieving up to **+5.4 BLEU** through SimulEval-based evaluation on ACL en-zh testset

Backend Development Intern | WeRide, Shanghai, CN

05/2025 - 08/2025

- Reduced Knative cold-start latency by 88% (12.2s to 1.5s) by building a service warming pool with symlink-based resource mapping and **Kubernetes** state recovery backed by PostgreSQL
- Productionized a containerized REST service with Bazel testing and CI/CD, and scaled **Prometheus/Grafana** health monitoring across a 120-GPU cluster, cutting scan time by 60% while maintaining 99.9% uptime

Machine Learning Engineer Intern | UMSN Pregnancy App, Ann Arbor, MI

05/2024 - 08/2024

- Led a 6-person team to launch a HIPAA-compliant AI prenatal care platform serving 20+ expectant mothers
- Integrated a DeepSeek-V3 RAG agent with n8n to generate personalized care summaries and nursing tasks; built Redis/JWT authentication that reduced login latency by 60% and database load by 70%

Software Development Engineer Intern | AVIAGE SYSTEMS, Shanghai, CN

05/2023 - 08/2023

- Built a Pandas-based flight-data pipeline for 13 sensor streams and optimized MySQL-backed REST APIs, reducing p95 query latency by 42% and database load by 55%

PROJECT EXPERIENCE

Sparse Attention on Blackwell | NVIDIA MLSys 2026 Competition

02/2026 - 04/2026

- Developed a Blackwell-optimized **CUDA** kernel for DeepSeek-V3 sparse attention, applying **FlashAttention**-style IO-aware fusion to reduce **KV-cache** traffic and accelerate long-context LLM inference
- Optimized Tensor Core execution, memory pipelining, parallel decomposition, and workload autotuning on **NVIDIA B200**; passed 23/23 attention workloads at ~57 us, achieving 22-35x speedup over the PyTorch reference

Hybrid Retrieval-Augmented Generation System | CMU Advanced NLP

01/2026 - 03/2026

- Built an end-to-end **RAG** pipeline from scratch, converting heterogeneous HTML and PDF sources into chunked embeddings and searchable FAISS/BM25 indexes
- Engineered hybrid retrieval with Reciprocal Rank Fusion and local open-source LLM inference for domain-specific factual QA, with automated evaluation across retrieval relevance and answer quality